Piper PA-31-350 Chieftain (CHIEF)

2× Lycoming TIO-540-J2BD / LTIO-540-J2BD turbocharged, counter-rotating, 350 hp each at 2575 RPM / 49 in Hg · 8–10 seats · Typical PA-31-350 Chieftain (7,000 lb MTOW, stretched unpressurised Navajo, 1973–1984) POH procedures; counter-rotating engines (no critical engine), hydraulic gear and flaps, inboard/outboard tanks per side, rear airstair plus cargo door. Unfeathering accumulators and emergency gear extension (CO2 or hand pump) vary by serial – verify your aircraft.

V-speeds & limits

Vr	**85–90 KIAS (rotate / liftoff)**
Vmca	**76 KIAS (red line)**
Vsse	**85 KIAS approx (intentional one-engine-inop)**
Vx	**85 KIAS**
Vxse	**95 KIAS**
Vyse	**106 KIAS (blue line)**
Vy	**110 KIAS**
Best glide	**approx 110–120 KIAS (both engines out, props feathered – verify POH)**
Vfe	**150 KIAS (15°) / 125 KIAS (40°)**
Vle	**156 KIAS (gear extended / extension)**
Vlo	**130 KIAS (retraction)**
Va	**158 KIAS (7,000 lb)**
Vno	**187 KIAS**
Vne	**236 KIAS**
Vs / Vso	**approx 76 / 68 KIAS (max weight, verify)**
Vref	**100 KIAS final (flaps 40°); 106 KIAS single-engine until landing assured**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel caps, drains, turbo exhaust / tailpipes, gear doors, oil 8–12 qt each)**
Fuel quantity	**CHECK visually – inboard 56 gal and outboard 40 gal per side (approx 182 gal usable total)**
Weight & balance	**CHECK (aft CG with full cabin and baggage – nose baggage, wing lockers, aft compartment)**
Nose baggage door / wing locker doors	**CLOSED & LOCKED**
Cargo door (aft of airstair)	**CLOSED & LOCKED – both lock pins engaged, handle flush, checked before airstair door is closed**
Passenger briefing	**COMPLETE (commuter brief: belts and seat backs, airstair and cargo door operation, overwing exit, no smoking, baggage stowed, briefing cards, oxygen if fitted)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Airstair cabin door	**CLOSED & LATCHED (upper and lower halves, handle flush, pins visible)**
Parking brake	**SET**
Landing gear selector	**DOWN**
Avionics master	**OFF**
All electrical switches	**OFF**
Circuit breakers	**CHECK IN**
Alternate static source	**OFF (normal)**
Fuel selectors	**INBOARD (both – takeoff and landing on inboard tanks)**
Crossfeed	**OFF**
Cowl flaps	**OPEN**
Alternator switches	**ON (both)**
Throttles / propellers / mixtures	**IDENTIFY – throttles closed, props full forward, mixtures idle cutoff**
Cabin heater	**OFF**

Engine Start

Battery master	**ON**
Gear lights	**THREE GREEN**
Annunciators	**PRESS TO TEST (DOOR annunciator out)**
Beacon	**ON**
Propeller area	**CLEAR (both sides)**
Mixture	**FULL RICH**
Propeller	**FULL FORWARD (high RPM)**
Throttle	**OPEN approx 1/2 inch**
Electric fuel pump	**ON until fuel flow indicates, then OFF (prime)**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Magneto / starter switch	**START (10 s max, then 30 s cool-down)**
Throttle	**800–1000 RPM – avoid rapid throttle movement with a cold turbocharger**
Oil pressure	**25 psi within 30 s, rising to 60–90 psi (100 psi max cold)**
Second engine	**REPEAT start sequence**
Alternators / ammeters	**CHECK both charging, voltage 27–29 V**
Electric fuel pumps	**OFF (fuel flow steady on engine-driven pumps)**
External power	**DISCONNECTED (if used)**
Avionics master	**ON**
Hydraulic pressure	**CHECK in green**
Engine instruments	**CHECK both (oil pressure, oil temp rising, fuel flow, TIT, CHT)**

Before Taxi

Flight instruments / avionics / GPS	**CHECK & SET**
Altimeters	**SET**
Heading indicator / HSI	**SET to compass**
Autopilot / electric trim	**CHECK & OFF**
Transponder	**STBY / as required**
Parking brake	**RELEASE**
Brakes	**CHECK on initial roll (heavier aircraft – allow for longer stopping)**
Nosewheel steering	**CHECK (rudder pedals)**
Gyros	**CHECK during turns (turn coordinator, DG, attitude indicator)**
Fuel selectors	**INBOARD both, crossfeed OFF**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET (into wind, oil temp above 75°F / 24°C before run-up)**
Flight controls	**FREE & CORRECT**
Trims (elevator, rudder, aileron)	**SET for takeoff**
Fuel selectors	**INBOARD both (feel for detents), crossfeed OFF**
Mixtures	**FULL RICH**
Propellers	**FULL FORWARD**
Throttles	**2000 RPM (one engine at a time)**
Magnetos	**CHECK L & R each engine (175 RPM max drop, 50 RPM max difference)**
Propellers	**CYCLE 2–3 times to low RPM (RPM drops approx 300–500, returns)**
Propellers – feather check	**1500 RPM, pull through detent to FEATHER – RPM drops; return to full forward before 1000 RPM**
Unfeathering accumulators	**CHECK charged (if fitted – allow pressure to rebuild after feather check)**
Alternators	**CHECK both charging under load**
Gyro suction / vacuum	**CHECK 4.8–5.1 in Hg, both pump annunciators out**
Engine instruments	**CHECK both (oil pressure 60–90 psi, oil temp green, CHT, TIT, fuel flow)**
Hydraulic pressure	**CHECK in green**
Throttles	**IDLE check (600–700 RPM), then 1000 RPM**
Electric fuel pumps	**ON (both)**
Cowl flaps	**OPEN**
Propellers	**FULL FORWARD**
Flaps	**UP or 15° (short field) – check indicator**
Landing gear selector	**DOWN – three green**
Pitot heat / de-ice	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Airstair door / cargo door / windows	**LATCHED (DOOR annunciator out)**
Takeoff briefing	**COMPLETE (Vmca 76 / Vr 85–90 / Vyse 106; abort below liftoff; engine-out plan – no critical engine, either failure handled the same)**
Takeoff power	**ADVANCE SMOOTHLY – 49 in Hg / 2575 RPM max, retard if MP overshoots (do not overboost)**

Turbocharger response lags the throttle. Advance throttles slowly over 3–5 seconds and check manifold pressure before brake release; set final power early in the roll.

After Takeoff / Climb

Landing gear	**UP (positive rate, no usable runway, below 130 KIAS)**
Gear lights	**OUT (three green out, no red in-transit light)**
Flaps	**UP (if used, above 95 KIAS)**
Climb power	**REDUCE SMOOTHLY – 2400 RPM / approx 38–40 in Hg (75%) after obstacles**
Airspeed	**110 KIAS (Vy) / 120–130 KIAS cruise climb**
Mixtures	**FULL RICH (fuel flow in green; lean only per POH climb schedule, TIT below 1650°F)**
Electric fuel pumps	**OFF above approx 1000 ft AGL (check fuel flow steady)**
Cowl flaps	**OPEN (CHT in green, oil temp below 245°F)**
Engine instruments	**CHECK both (MP, RPM, fuel flow, TIT, CHT, oil)**
Landing / taxi lights	**OFF**

Cruise

Cruise power	**SET smoothly – 2200–2400 RPM / 32–36 in Hg typical (set MP after RPM)**
Mixtures	**LEAN to TIT per POH cruise table (1650°F max, typically 100°F rich of peak)**
Cowl flaps	**CLOSED (as required for CHT)**
Propellers	**SYNCHRONISE**
Fuel selectors	**OUTBOARD (level cruise only) – electric fuel pumps ON during tank change, OFF when flow steady**
Fuel quantity / flow	**CHECK against plan; return to INBOARD before outboards run dry and for descent**
Crossfeed	**OFF (one-engine-inop balance only)**
Engine instruments	**CHECK both (TIT, CHT, oil temp 245°F max, oil pressure 60–90 psi)**
Altimeter / heading	**CHECK**
Cabin heater / oxygen	**AS REQUIRED**

Descent

Altimeters	**SET**
Fuel selectors	**INBOARD both – electric fuel pumps ON during change, then OFF**
Power	**REDUCE GRADUALLY (approx 2 in Hg per minute – avoid shock cooling; keep MP above gear horn setting)**
Mixtures	**ENRICH progressively**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 156 KIAS for gear, below 150 KIAS for 15° flap**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Seat belts / harnesses	**SECURE – passengers briefed, seat backs upright, cabin secure**

Before Landing

Fuel selectors	**INBOARD both, crossfeed OFF**
Electric fuel pumps	**ON (both)**
Mixtures	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 156 KIAS**
Landing gear	**DOWN – THREE GREEN, hydraulic pressure recovered**
Flaps	**15° below 150 KIAS; 40° on final below 125 KIAS**
Propellers	**FULL FORWARD (on final)**
Airspeed	**100 KIAS final; approx 90–95 KIAS over threshold**
Landing lights	**ON**
Landing gear	**THREE GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (identify flap lever before moving)**
Cowl flaps	**OPEN**
Electric fuel pumps	**OFF**
Pitot heat / de-ice	**OFF**
Landing lights / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttles	**1000 RPM for 3–5 min (turbocharger cool-down before shutdown)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Mixtures	**IDLE CUTOFF**
Magnetos	**OFF (after propellers stop)**
Alternator switches / battery master	**OFF**
Fuel selectors	**OFF (if required by operator) or INBOARD**
Control lock	**INSTALL**
Passengers	**DEPLANE via airstair only after props stopped; cargo door as required for baggage**
Chocks / tiedowns / pitot covers	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT annunciator, one ammeter zero or voltage below approx 26 V

Affected alternator switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

▶ Alternator restored?

Yes:

Ammeters / voltmeter ..... **MONITOR both**

No – one alternator failed:

Affected alternator switch ..... **OFF**

Electrical load ..... **REDUCE to within remaining alternator capacity (100 A – de-ice and heater loads first)**

Deferred: One alternator inoperative – electrical load reduced, land as soon as practical. ..... **NOTED**

No – both failed:

Both alternator switches ..... **OFF**

Non-essential electrical load ..... **REDUCE (battery only, limited endurance; avionics minimum)**

Landing ..... **PLAN early – gear is hydraulic but gear lights, electric fuel pumps and flap indication need battery power**

Deferred: Both alternators failed – battery only; land as soon as possible, confirm gear by hydraulic pressure cycle. ..... **NOTED**

Low Oil Pressure (one engine)

Oil pressure below 60 psi in flight or below 25 psi at idle

Oil pressure / oil temperature ..... **CHECK and cross-check the affected engine**

▶ Oil temperature rising or pressure below 25 psi?

Yes – genuine:

Affected engine power ..... **REDUCE to minimum required**

Engine ..... **SHUT DOWN and FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**

Feathering needs oil pressure – feather early while the engine still turns.

Deferred: Low oil pressure – precautionary engine shutdown; single-engine landing, land as soon as possible. ..... **NOTED**

No – gauge suspect:

Flight ..... **CONTINUE, monitor oil temperature and pressure**

Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

Turbocharger Failure / Manifold Pressure Loss

Manifold pressure drops toward ambient and cannot be restored; possible exhaust leak, smoke or roughness

Affected engine throttle ..... **REDUCE, then re-set smoothly – do not pump the throttle**

Affected engine mixture ..... **LEAN to fuel flow appropriate to the reduced power (TIT / EGT rises as mixture richens on a dead turbo)**

Engine instruments ..... **CHECK oil pressure, oil temp, TIT, CHT, fuel flow**

▶ Engine smooth with normally-aspirated power, no smoke or fire indication?

Yes:

Power ..... **OPERATE at the reduced power available; mixture as required**

Altitude ..... **DESCEND as required – single-engine ceiling and cruise altitude reduced**

Deferred: Turbocharger inoperative one side – reduced power and asymmetric thrust at altitude, land as soon as practical. ..... **NOTED**

No – roughness, smoke or fluctuating oil pressure:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight) – exhaust leak may cause nacelle fire**

Deferred: Engine shut down for turbo / exhaust failure – single-engine landing, land as soon as possible. ..... **NOTED**

High TIT / Overboost

TIT above 1650°F or manifold pressure above 49 in Hg

Affected engine throttle ..... **RETARD to bring MP below 49 in Hg**

Affected engine mixture ..... **ENRICH until TIT below 1650°F**

Affected engine cowl flap ..... **OPEN**

Airspeed ..... **INCREASE if CHT also high**

▶ TIT and MP within limits?

Yes:

Power / mixture ..... **RE-SET slowly; note any overboost (duration and MP) for maintenance**

Deferred: Overboost or TIT exceedance – record for maintenance entry. ..... **NOTED**

No:

Affected engine power ..... **REDUCE further – consider shutdown if TIT uncontrollable (Engine Failure in Flight)**

Deferred: TIT / MP uncontrollable – wastegate or controller fault, land as soon as practical. ..... **NOTED**

Fuel Flow Fluctuation / Vapour / Rough Running

Affected engine electric fuel pump ..... **ON**

Fuel selector ..... **CHECK INBOARD (or tank with fuel) – outboard tanks unport in turns or with low fuel**

Affected engine mixture ..... **ADJUST for smooth running**

Affected engine magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (fuel flow, TIT, CHT, MP)**

▶ Engine running smoothly?

Yes:

Electric fuel pump ..... **LEAVE ON for the remainder of flight**

Deferred: Fuel flow fluctuation – electric pump on, inboard tanks; report to maintenance. ..... **NOTED**

No:

Crossfeed ..... **ON from other side if fuel starvation suspected**

Affected engine power ..... **REDUCE to smoothest setting – feather if power unusable (Engine Failure in Flight)**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Crossfeed Operation

Fuel balance with one engine inoperative, or fuel starvation suspected on one side

Flight attitude ..... **LEVEL (crossfeed is for level flight only)**

Fuel selector – operating engine ..... **INBOARD or OUTBOARD as required to select the tank supplying**

Fuel selector – inoperative engine side ..... **ON tank to be used (INBOARD or OUTBOARD)**

Crossfeed ..... **ON**

Electric fuel pumps ..... **ON during changeover, then as required**

Fuel flow / quantity ..... **MONITOR both sides**

Crossfeed ..... **OFF before approach – operating engine on its own INBOARD tank for landing**

Do not take off or land with crossfeed ON.

Deferred: Crossfeed used in flight – confirm crossfeed OFF and inboard tank selected before landing. ..... **NOTED**

Propeller Overspeed

RPM above 2575 and not responding to the prop lever

Affected engine throttle ..... **REDUCE**

Affected engine propeller ..... **LOW RPM (toward feather detent)**

Airspeed ..... **REDUCE to bring RPM within limits**

Affected engine oil pressure ..... **CHECK (governor is oil-operated)**

▶ RPM controllable at or below 2575?

Yes:

Power ..... **SET minimum that keeps RPM in limits**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

No:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight)**

Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. ..... **NOTED**

Flap Failure / Asymmetry

Flap lever ..... **RETURN to current position – do not continue extension**

Hydraulic pressure ..... **CHECK (flaps and gear share the hydraulic system)**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – flaps up approach approx 110–115 KIAS, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance; check gear extends normally. ..... **NOTED**

Gear Unsafe / Fails to Retract or Extend Normally

Red in-transit light stays on, fewer than three green, or gear warning horn

Airspeed ..... **BELOW 156 KIAS (below 130 KIAS for retraction)**

Hydraulic pressure ..... **CHECK – allow time for pump to complete cycle**

Gear light bulbs / navigation lights dimmer ..... **CHECK (bulb or dimming, swap bulbs)**

Landing gear selector ..... **RECYCLE once to the desired position**

▶ Gear in desired position with correct indication?

Yes:

Gear indication ..... **CONFIRM**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear selector ..... **DOWN – leave down**

Airspeed ..... **BELOW 156 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range (negative single-engine climb), land as soon as practical. ..... **NOTED**

No – failed to extend / unsafe:

Use Emergency Gear Extension. Request a visual check from ATC or another aircraft if available.

Airstair / Cargo Door Open in Flight

DOOR annunciator, noise or draught from the aft cabin

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110–120 KIAS**

Belts / loose items / passengers ..... **SECURE, passengers moved forward away from the doors**

▶ Which door?

Airstair door:

Do not attempt to close the airstair door in flight. The lower (stair) half may trail open; expect noise and buffet.

Cargo door:

The large cargo door can depart the aircraft or strike the tail if it opens fully. Fly smoothly, avoid sideslip toward the open door, and land at the nearest suitable airport.

Baggage in aft compartment ..... **CHECK secured – loose items may exit through the door**

Deferred: Cabin / cargo door unlatched – return and land as soon as practical, normal approach speed, door and hinge inspection. ..... **NOTED**

Vacuum / Gyro Failure

Vacuum annunciator on, suction below 4.8 in Hg, or attitude / heading indicator drifting

Suction gauge / pump annunciators ..... **CHECK (one engine-driven pump failed – remaining pump maintains system)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**

Failed instruments ..... **COVER**

Autopilot ..... **OFF if coupled to failed gyro**

IMC ..... **EXIT / AVOID if both pumps or the attitude indicator failed**

Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach. ..... **NOTED**

Inadvertent Icing Encounter

Ice accumulating on wings, windshield or props

Pitot heat / stall warning heat ..... **ON**

Propeller de-ice / windshield heat ..... **ON (if fitted)**

Wing / tail de-ice boots ..... **CYCLE when approx 1/2 inch of ice has formed (if fitted)**

Alternate air ..... **CHECK – induction icing shows as MP loss; alternate air is automatic on the TIO-540**

Power ..... **INCREASE as required; cowl flaps as required**

Altitude / heading ..... **CHANGE to exit icing conditions (ask ATC)**

Autopilot ..... **OFF – hand fly, monitor airspeed**

▶ Aircraft certified and equipped for flight in known icing?

Yes:

Approach ..... **FLAPS 15° max, speed +10–15 KIAS, no slow flight**

Deferred: Ice accretion – partial-flap landing, approach speed increased, longer runway. ..... **NOTED**

No:

Icing conditions ..... **EXIT IMMEDIATELY**

Deferred: Icing encounter without approved equipment – flaps 15° max, approach speed increased, land as soon as practical. ..... **NOTED**

Pitot / Static Blockage

Airspeed, altimeter or VSI erratic or frozen

Pitot heat ..... **ON**

Alternate static source ..... **ON (pilot subpanel – expect small altimeter and airspeed changes)**

Cabin / storm windows / vents ..... **CLOSED when using alternate static**

Attitude / power / GPS groundspeed ..... **USE for pitch and speed reference**

▶ **Indications restored?**

Yes:

Deferred: Alternate static in use – apply POH correction, pitot / static drain and inspection required. ..... **NOTED**

No:

Approach ..... **FLY by attitude and power (known settings), VMC preferred, longer runway**

Deferred: Pitot / static blocked – unreliable airspeed, fly known power and attitude settings. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff

On the runway or just airborne with runway remaining

▶ **Airborne?**

No – below liftoff:

Throttles (memory) ..... **CLOSED (both)**

Brakes (memory) ..... **MAXIMUM – stop straight ahead**

Mixtures ..... **IDLE CUTOFF**

Fuel selectors ..... **OFF**

Magnetos ..... **OFF**

Battery master / alternators ..... **OFF**

Yes – runway remaining:

Throttles (memory) ..... **CLOSED (both)**

Land (memory) ..... **STRAIGHT AHEAD, gear down**

Brakes ..... **MAXIMUM**

Mixtures / fuel selectors / magnetos ..... **OFF after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (76 KIAS) directional control cannot be maintained with full power on the operating engine. Do not attempt to continue a takeoff below liftoff speed. Counter-rotating engines: neither engine is critical, but Vmca still applies.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

Mixtures (memory) ..... **FULL RICH**

Propellers (memory) ..... **FULL FORWARD**

Throttles (memory) ..... **FULL (49 in Hg max)**

Flaps (memory) ..... **UP**

Airspeed (memory) ..... **106 KIAS (Vyse) – never below 76 KIAS**

▶ **Landing gear position?**

Gear DOWN:

With gear down single-engine climb is marginal or negative at most weights, especially near 7,000 lb. Retract if committed to fly; otherwise land straight ahead.

Landing gear (memory) ..... **UP (if committed to fly)**

Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**

Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**

Inoperative engine propeller (memory) ..... **FEATHER**

Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

Gear UP:

Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**

Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**

Inoperative engine propeller (memory) ..... **FEATHER**

Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

Bank ..... **2–5° toward operating engine, ball half out**

Trim ..... **RUDDER & AILERON for straight flight**

Climb ..... **106 KIAS to safe altitude, then Engine Securing**

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. ..... **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

- Mixtures (memory) ..... **FULL RICH**
- Propellers (memory) ..... **FULL FORWARD**
- Throttles (memory) ..... **FULL (as required, 49 in Hg max)**
- Landing gear / flaps (memory) ..... **UP**
- Airspeed (memory) ..... **106 KIAS (Vyse) minimum**
- Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**
- Inoperative engine propeller (memory) ..... **FEATHER**
- Inoperative engine mixture (memory) ..... **IDLE CUTOFF**
- Bank / rudder ..... **2–5° toward operating engine, ball half out**
- Trim ..... **RUDDER & AILERON**

▶ Cause obvious (fuel / tank empty) and altitude available for a restart?

Yes – attempt air start:

- Fuel selector ..... **INBOARD (or tank with fuel), electric fuel pump ON**

Use the Air Start procedure. If unsuccessful, secure the engine.

No – secure engine:

Continue with Engine Securing, then divert to the nearest suitable airport. Single-engine service ceiling is low at high weight – check drift-down altitude against terrain.

Deferred: One engine inoperative – single-engine approach and landing, drift-down altitude check, no go-around once committed. ..... **NOTED**

Engine Securing (after feather)

Inoperative engine propeller feathered

- Inoperative engine fuel selector ..... **OFF**
- Inoperative engine electric fuel pump ..... **OFF**
- Inoperative engine magnetos ..... **OFF**
- Inoperative engine alternator ..... **OFF**
- Inoperative engine cowl flap ..... **CLOSED**
- Propeller synchrophaser ..... **OFF (if fitted)**
- Electrical load ..... **REDUCE (single alternator)**
- Operating engine ..... **MONITOR – power as required (49 in Hg max), cowl flap as required, mixture adjusted, TIT below 1650°F**
- Crossfeed ..... **ON as required to use fuel from the dead-engine side (level flight only; both electric pumps as required)**
- Trim ..... **ADJUST for straight flight**
- Passengers ..... **BRIEF – remain seated, belts tight**
- ATC ..... **ADVISE – nearest suitable airport**

Hydraulic pressure for gear and flaps is available from the operating engine pump; expect slower gear and flap operation.

Air Start (Unfeather)

Restart attempt with altitude and cause identified

- Fuel selector ..... **INBOARD (tank with fuel)**
- Electric fuel pump ..... **ON**
- Throttle ..... **OPEN approx 1/4**
- Mixture ..... **FULL RICH**
- Magnetos ..... **ON (both)**
- Alternator ..... **OFF until running**
- Propeller ..... **FORWARD out of feather detent to mid-range (accumulator unfeathers if fitted – allow 10–20 s)**
- Starter ..... **ENGAGE if propeller does not windmill (airspeed 100–120 KIAS helps)**

▶ Engine restarted?

Yes:

- Throttle ..... **1000–1500 RPM until oil pressure and temperature normal, then increase power SLOWLY (cold turbocharger)**
- Alternator ..... **ON**
- Electric fuel pump ..... **OFF when fuel flow steady**
- Mixture / cowl flap ..... **AS REQUIRED**

Deferred: Engine restarted in flight – land as soon as practical, cause investigation required. ..... **NOTED**

No:

- Propeller ..... **FEATHER**
- Mixture ..... **IDLE CUTOFF**

Complete Engine Securing.

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel selector	**INBOARD (crossfeed as required until final approach, then OFF)**
Operating engine electric fuel pump	**ON**
Mixture	**FULL RICH**
Propeller	**FULL FORWARD**
Cowl flap	**AS REQUIRED**
Airspeed	**106 KIAS (Vyse) until landing assured**
Approach	**NORMAL glidepath or slightly high – avoid low, flat approaches and large power changes**
Landing gear	**DOWN when landing assured (within gliding distance) – allow for slower hydraulic extension**
Flaps	**15° when landing assured; 40° on short final only**
Airspeed	**REDUCE below 106 KIAS only on short final**

▶ Go-around required?

No:

Landing	**NORMAL – brakes as required, expect yaw as power is reduced**

Yes – above decision height:

Operating engine	**FULL POWER (smoothly, 49 in Hg max)**
Flaps	**UP**
Landing gear	**UP (positive climb)**
Airspeed	**106 KIAS (Vyse)**
Bank	**2–5° toward operating engine**

A single-engine go-around below approx 300 ft AGL, or once 40° flap is extended, should not be attempted – at high weight the aircraft may not climb.

Engine Fire in Flight

LAND IMMEDIATELY

Affected engine fuel selector (memory)	**OFF**
Affected engine throttle (memory)	**CLOSED**
Affected engine propeller (memory)	**FEATHER**
Affected engine mixture (memory)	**IDLE CUTOFF**
Affected engine electric fuel pump (memory)	**OFF**
Cabin heater / defroster	**OFF**
Affected engine magnetos	**OFF**
Affected engine alternator	**OFF**
Affected engine cowl flap	**CLOSED**
Crossfeed	**OFF**
Airspeed	**106 KIAS or higher to help extinguish fire (descend if required)**
Landing	**NEAREST SUITABLE AIRPORT – do not attempt restart**

Engine Fire During Start

Starter (memory)	**CONTINUE CRANKING**

▶ Engine starts?

Yes:

Throttle	**1500–1700 RPM for a few minutes**
Engine	**SHUT DOWN and inspect for damage**

No:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Electric fuel pump	**OFF**
Fuel selector	**OFF**
Starter	**CONTINUE CRANKING to draw fire into engine**
Magnetos / battery master / alternators	**OFF**
Fire extinguisher	**OBTAIN / USE**
Aircraft	**EVACUATE passengers via airstair (cargo door if airstair blocked)**

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

Battery master / alternator switches (memory)	**OFF**
Avionics master (memory)	**OFF**
Cabin heater (memory)	**OFF**
Cabin vents / air	**CLOSED until fire out, then OPEN to ventilate**
Fire extinguisher	**ACTIVATE (then ventilate cabin)**
Passengers	**BRIEF – oxygen if smoke, belts tight**

▶ Fire out and electrical power required?

Yes:

Circuit breakers	**CHECK – do not reset faulted**
Battery master	**ON**
Alternators / avionics / equipment	**ON one at a time, check for smoke**
Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable.	**NOTED**

No:

Gear and flaps are hydraulic and remain available. Gear position lights, fuel quantity, electric fuel pumps and gear warning are electric – confirm gear by hydraulic pressure cycle and visual check.

Deferred: Electrical power off – no gear lights or electric fuel pumps; land on inboard tanks, longer runway.	**NOTED**

Emergency Descent

Throttles (memory)	**CLOSED (slowly enough to avoid turbo surge; some MP helps cooling)**
Propellers (memory)	**FULL FORWARD**
Landing gear (memory)	**DOWN (below 156 KIAS)**
Flaps (memory)	**UP**
Airspeed (memory)	**156 KIAS (Vle) – smooth air; Vno 187 KIAS gear up if structural cooling allows**
Mixtures	**ENRICH as required**
Cowl flaps	**CLOSED**
Level-off	**POWER UP GRADUALLY – avoid rapid throttle advance**

Emergency Gear Extension

Gear does not extend with the selector; hydraulic pressure low or lost

Airspeed	**100 KIAS or less (slower is better)**
Landing gear selector	**DOWN**
Hydraulic pressure	**CHECK – if available, operating-engine pump may still extend gear; allow time**
Emergency gear extension lever / knob	**PULL and HOLD to the stop (floor, forward of pilot seat / between seats – releases uplocks; CO2 bottle assists on early aircraft)**
Hand pump	**OPERATE until three green (hand-pump variant, if fitted)**
Aircraft	**YAW gently side to side to help gear lock down**
Gear lights	**THREE GREEN**

Do not move the gear selector up or attempt to retract after an emergency extension. Leave the gear down.

▶ Three green obtained?

Yes:

Deferred: Gear extended by emergency system – do not retract; gear and hydraulic inspection required.	**NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Fuel	**BURN DOWN to minimum practical, land on inboard tanks**
Passenger briefing	**COMPLETE (belts tight, brace, airstair door, cargo door and overwing exit, baggage secured)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**40°**
Approach	**NORMAL, 95–100 KIAS**
Electric fuel pumps	**OFF**
Airstair door	**UNLATCHED on short final**

On short final:

Mixtures	**IDLE CUTOFF**
Fuel selectors	**OFF**
Magnetos	**OFF**
Battery master / alternators	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped – airstair, cargo door, overwing exit**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist cb0c6525 · 2026-11-07